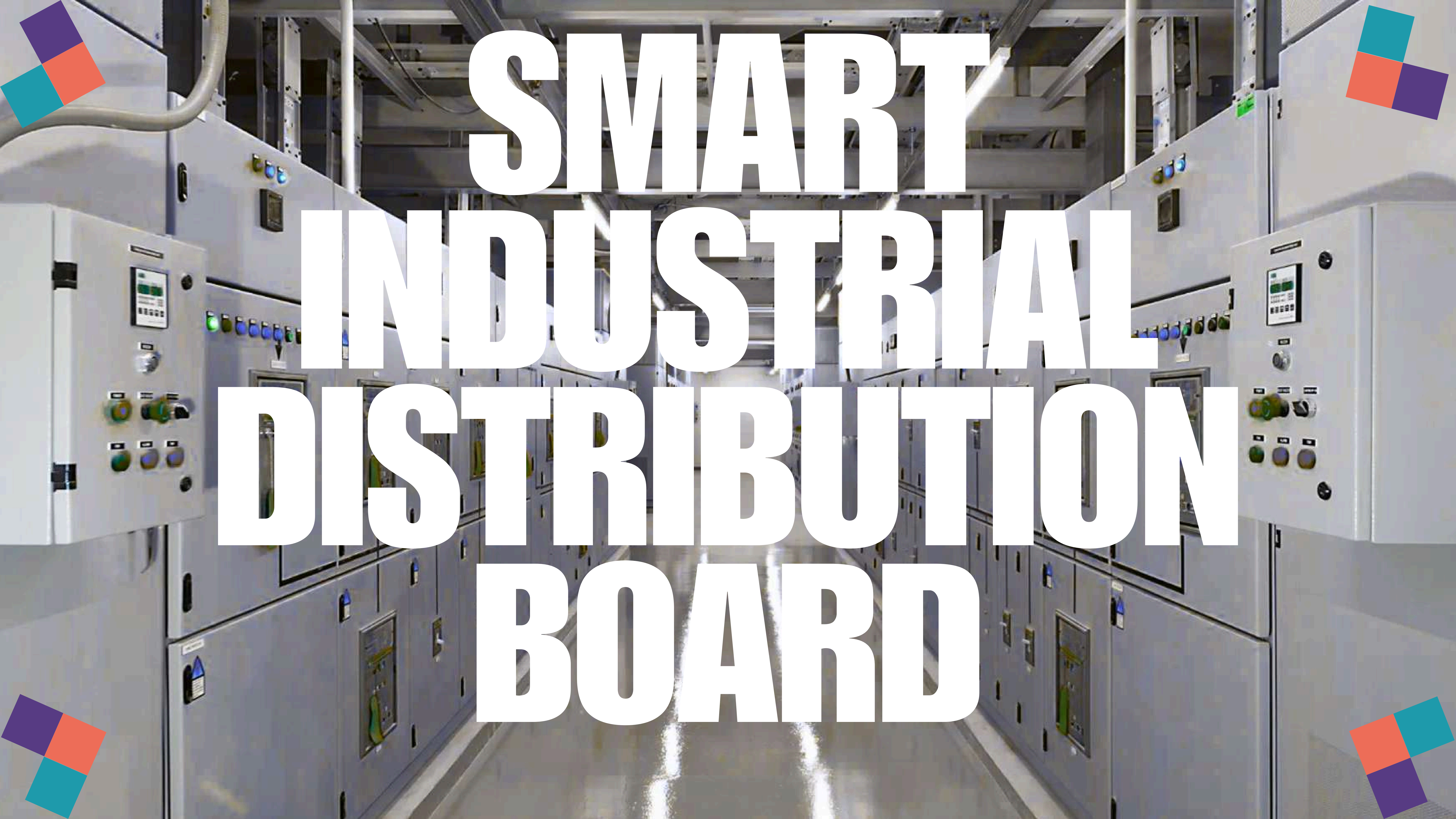

SMART INDUSTRIAL DISTRIBUTION BOARD

Agenda

- The team
- Introduction
- Problem Statement
- Proposed Solution
- Technical Design & Implementation
- Results & Analysis
- Project timeline

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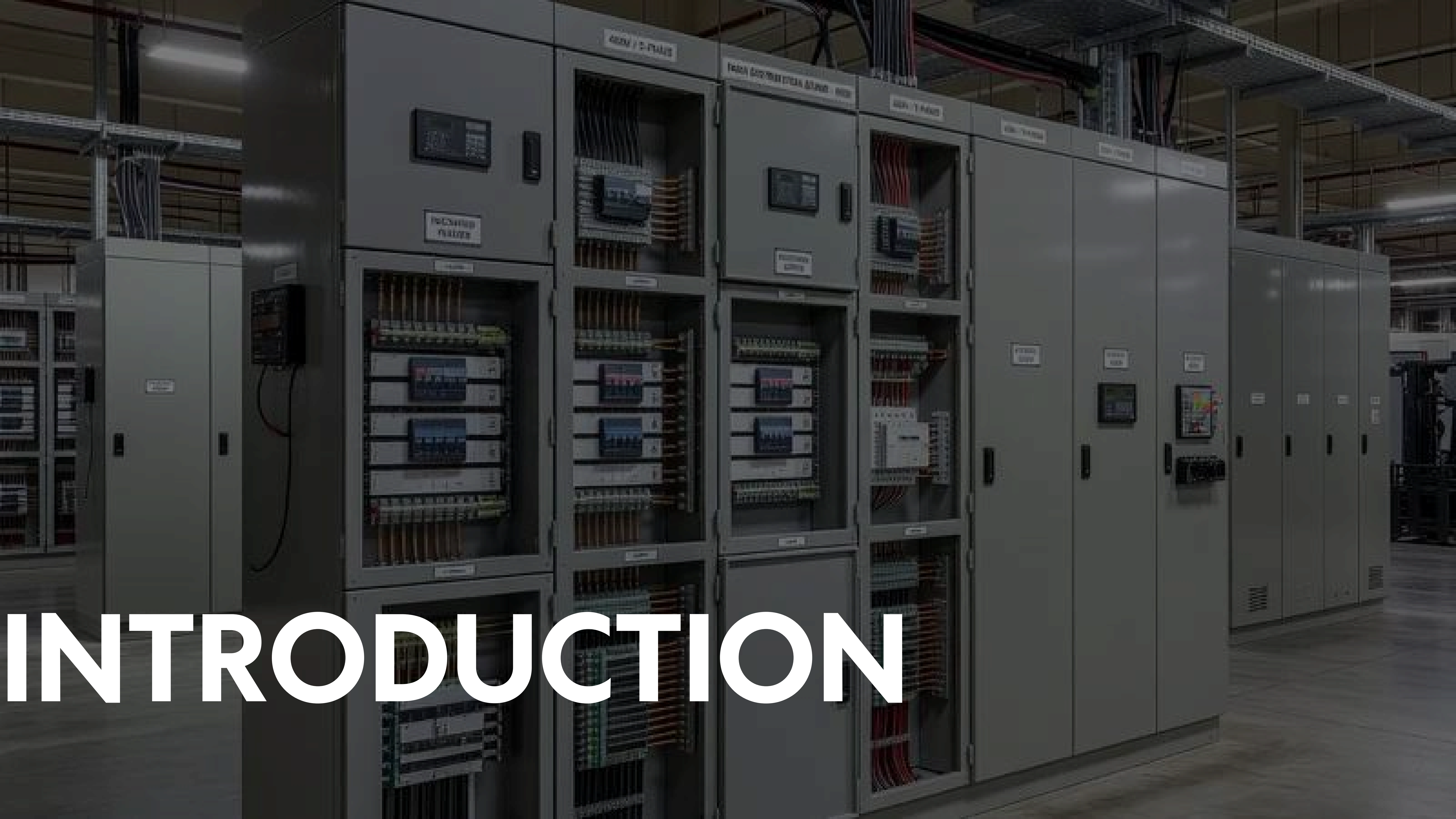
PROJECT OVERVIEW

We built an Arduino-based Smart Distribution Board providing 3-in-1 power protection with a black-box failure logger. It eliminates sensor noise and display flickering, offering a cost-effective alternative to expensive commercial equipment. This directly benefits maintenance technicians by simplifying troubleshooting and minimizing factory downtime.

SPRING 2026

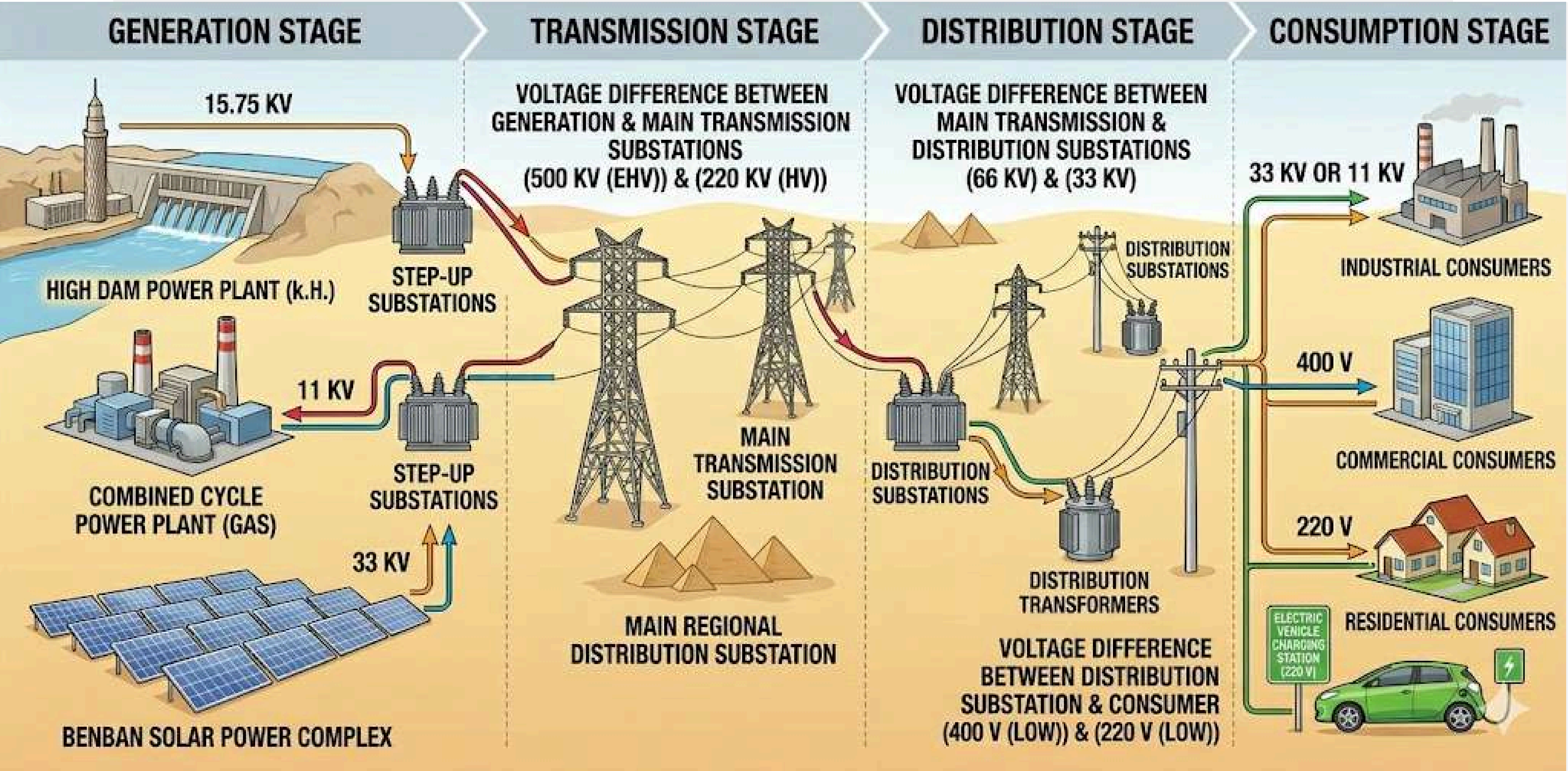


INTRODUCTION

POWER SYSTEM

An electric power system is a large-scale network comprising generation, transmission, and distribution components designed to produce, transport, and deliver electrical energy from suppliers to consumers. It operates in real-time, balancing generation with demand to power industrial, commercial, and residential loads. Key components include generators, transformers, and transmission lines.

POWER SYSTEM STRUCTURE



Generation

This is where electricity is produced by converting mechanical energy (using turbines) or chemical energy (using solar cells) into electrical energy. Common sources include:

- Conventional: Natural Gas, Nuclear, and Coal.
- Renewable: Solar, Wind, and Hydroelectric.

Transmission

Since power plants are often located far from urban centers, electricity must be transported over long distances. To minimize energy loss due to resistance Step-up Transformers increase the voltage to very high levels (e.g., 220kV or 500kV). High voltage allows for lower current, which significantly reduces heat loss during transit.

Distribution

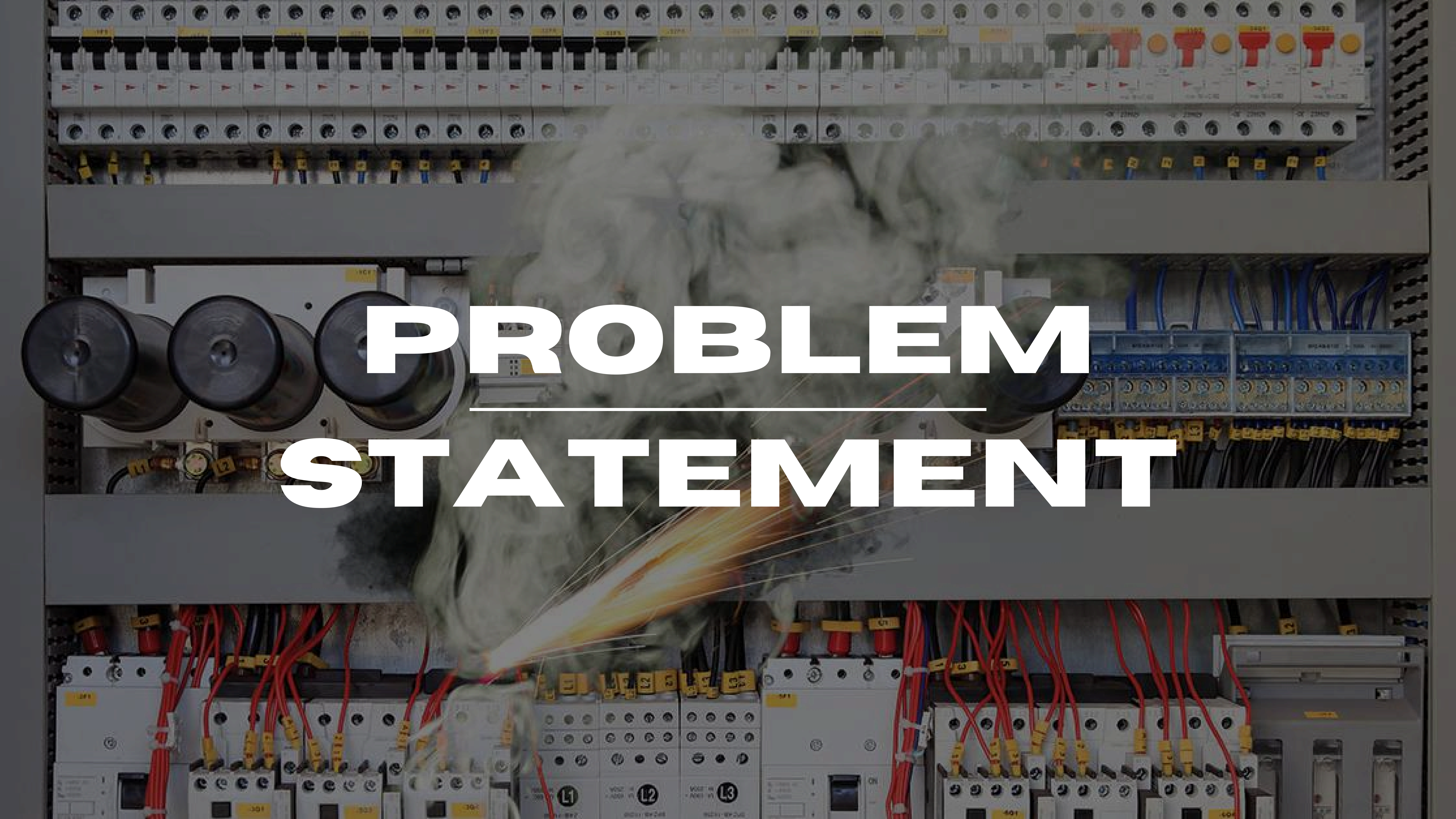
Once the power reaches the outskirts of a city or industrial zone, it enters the distribution phase. Step-down Transformers reduce the voltage to safer, usable levels. This stage connects the high-voltage transmission grid to the end consumers via medium and low-voltage lines.



Loads

The final stage where electrical energy is converted back into useful work, such as:

- Residential: Lighting, heating, and appliances.
- Industrial: Large induction motors and manufacturing equipment.
- Commercial: Office buildings and shopping malls.

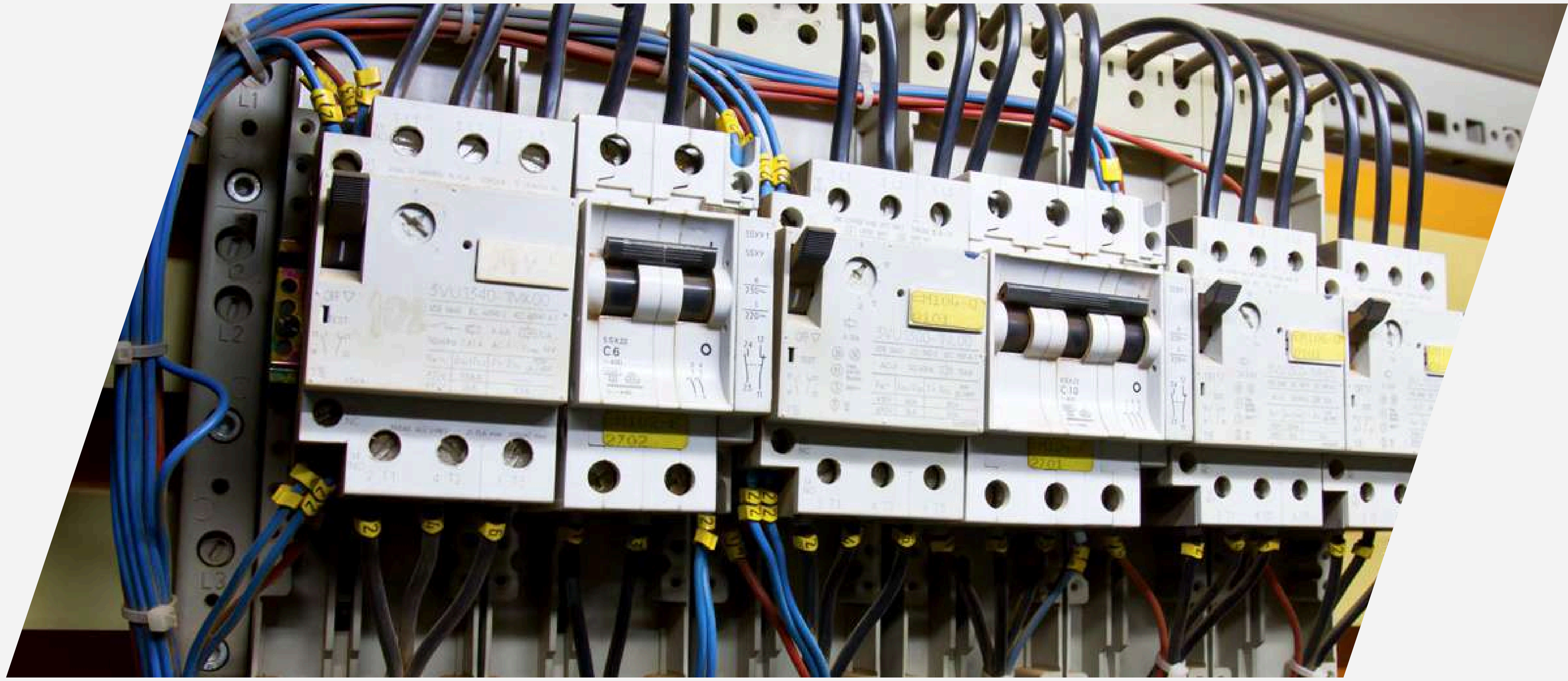


PROBLEM STATEMENT

THE DISADVANTAGES IN THE TRADIONAL DISTRIBTION BOARD

Traditional electrical distribution boards (DBs)—while reliable and foundational to modern infrastructure—suffer from several distinct disadvantages, especially when compared to modern smart distribution boards. Their limitations primarily stem from one major factor: they are entirely passive, "dumb" hardware.

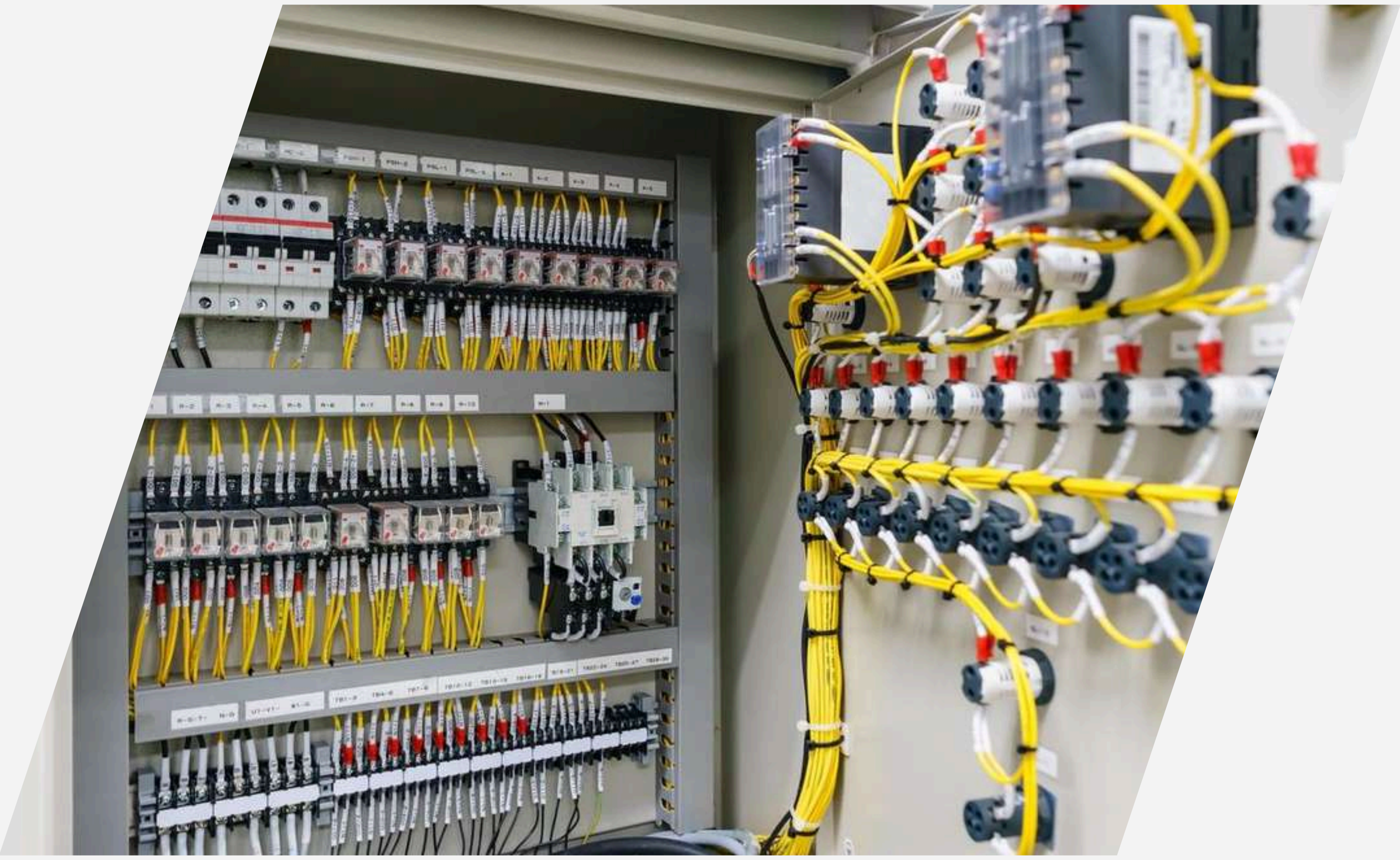
LACK OF REAL-TIME MONITORING AND DATA



Traditional distribution boards cannot interface with external software or networks.

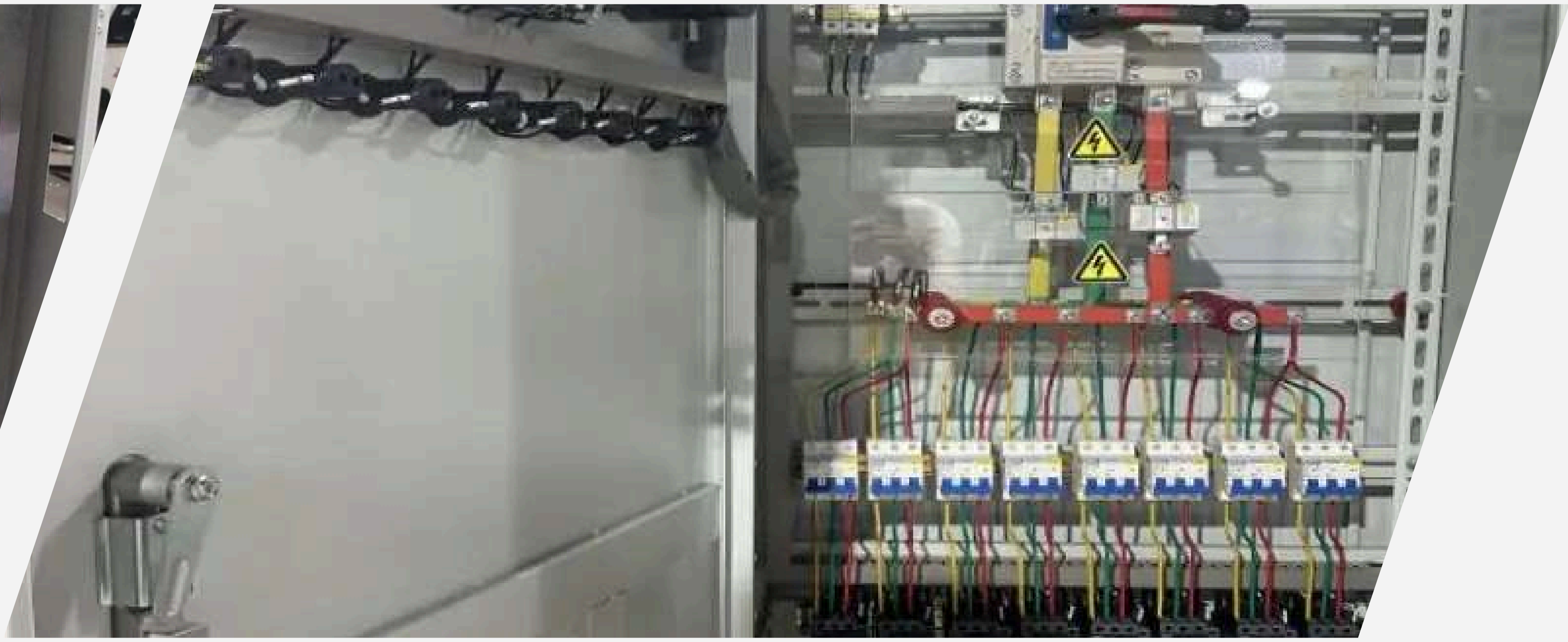
- Local Operations Only: You cannot remotely isolate a circuit, shed non-essential loads during a peak demand period, or cut power in an emergency unless you are standing right in front of the panel.
- Zero Automation Integration: They cannot naturally communicate with Building Management Systems (BMS), Supervisory Control and Data Acquisition (SCADA) systems, or industrial microgrids without extensive, expensive retrofitting.

REACTIVE MAINTENANCE



With a traditional board, you only find out there is a problem after a failure occurs.

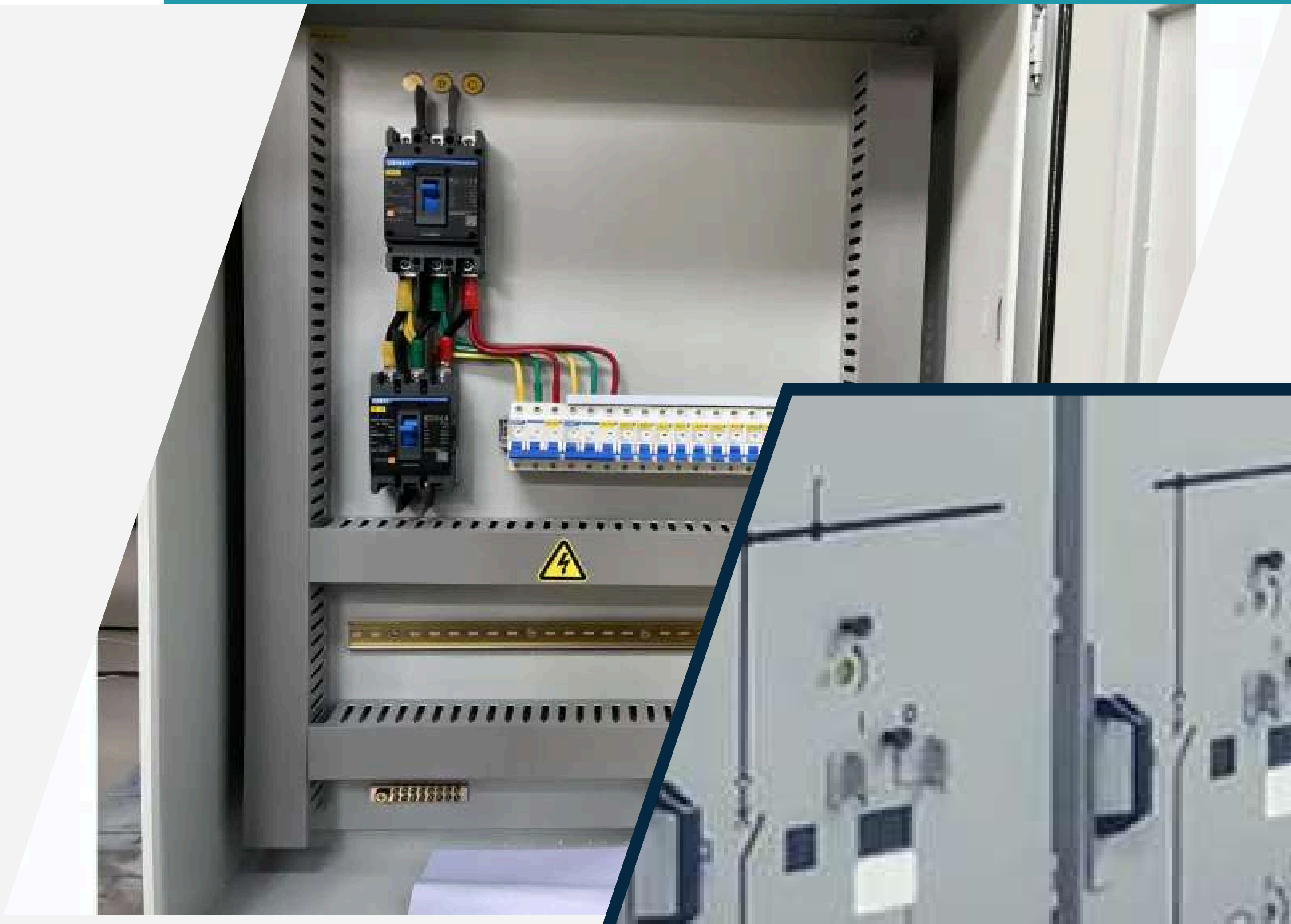
- Fixing After the Fact: If a breaker trips or a component overheats, you only notice when the power goes out or equipment shuts down.
- Hidden Failures: Minor faults, micro-arcs, or gradual insulation degradation go unnoticed until they escalate into catastrophic failures, equipment damage, or electrical fires.



HIGH DOWNTIME AND MANUAL TROUBLESHOOTING

When a fault occurs in a traditional setup, restoring power requires physical intervention.

- Manual Resetting: A maintenance technician must physically walk to the board, locate the tripped breaker, and flip it back on.
- Complex Diagnostics: Finding why a breaker tripped involves manual testing with multi-meters, which takes time and extends operational downtime in industrial settings.



NO REMOTE CONTROL OR AUTOMATION

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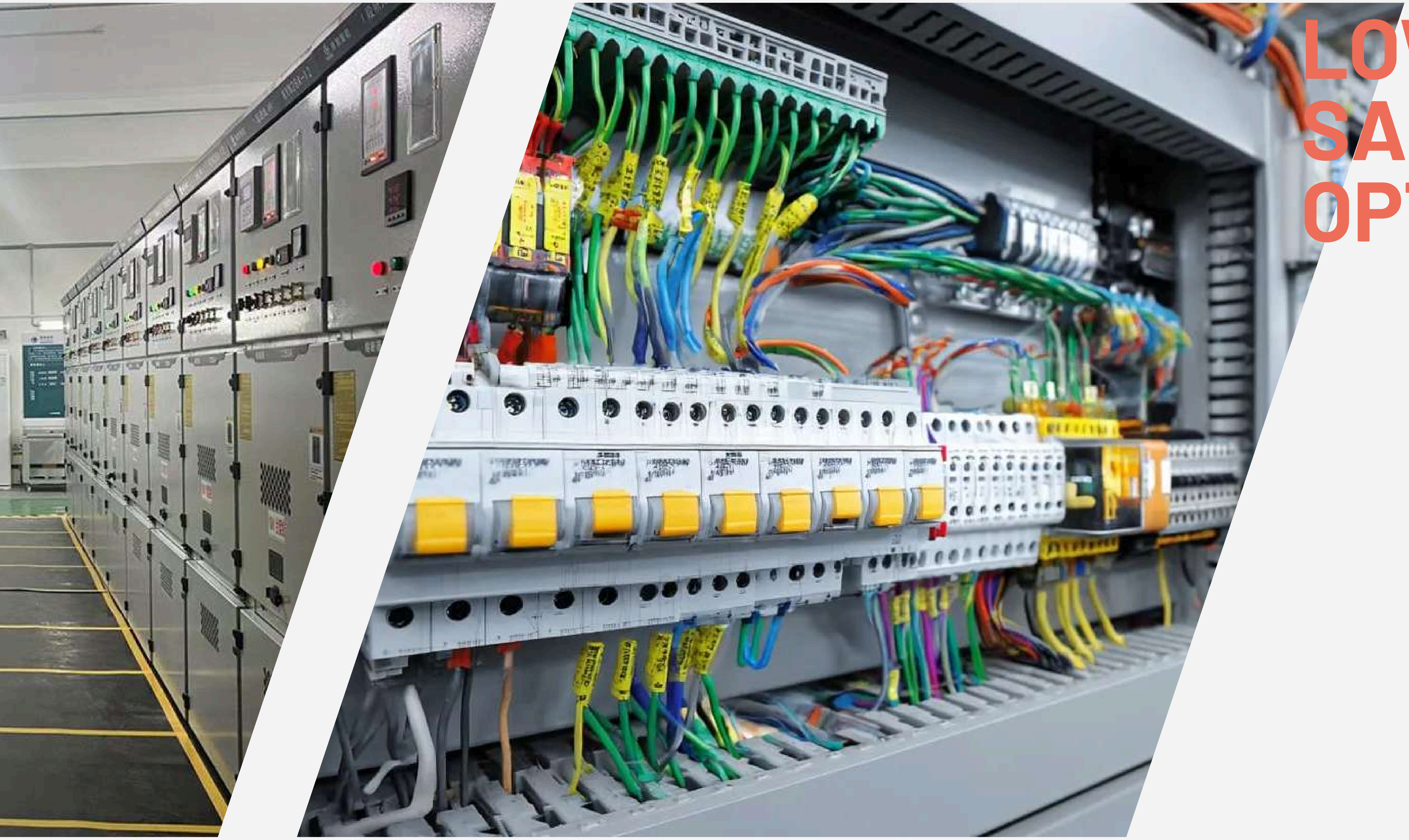
SMART INDUSTRIAL DISTRIBUTION BOARD



LOWER SAFETY OPTIMIZATION

While traditional circuit breakers (MCBs, MCCBs) protect against overcurrent and short circuits, their safety features are purely mechanical and thermal.

- Slower Reaction to Complex Faults: They are less sensitive to subtle, dangerous anomalies like low-level arc faults or transient voltage surges unless specific, separate protective devices are manually added.
- No Early Warning Systems: They cannot send a warning alert when a circuit is approaching 95% of its rated capacity; they simply trip when it's too late.

PROPOSED SOLUTION

PROPOSED SOLUTION

1

Intelligence at the Edge: Replaces standard distribution panels with an IoT-ready, microcontroller-driven hub.

2

Predictive & Proactive Safety: Monitors power anomalies (voltage spikes, minor overcurrents) to mitigate damage before traditional circuit breakers trip.

3

Integrated Human-Machine Interface (HMI): Provides clear local monitoring and physical control override via an intuitive, color-coded front panel

4

Modular Automation: Designed to seamlessly integrate into existing industrial frameworks to modernize factory floors without full-scale overhauls.

Our Approach

Dual-Layer Defense Architecture

We implemented a hybrid safety system. Layer 1 is digital and predictive (Sensors + Arduino monitoring thresholds). Layer 2 is mechanical and absolute (Molded Circuit Breakers acting as the final line of defense).

State-Machine Control Logic

The Arduino software is programmed with precise control loops that continuous read telemetry, update the HMI screen, and toggle relays dynamically based on real-time loads.

Noise Reduction & Isolation

As detailed in the internal wiring we strategically separated the low-voltage control circuitry (5V/ 2V DC) from the high-voltage power lines (220 V/380V AC) to eliminate Electromagnetic Interference (EMI).

Thermal Optimization

Enclosure physics were taken into account by integrating an active cooling loop (intake regulator and exhaust fan) to prevent component degradation from localized heat buildup.tters.

Key Features

1

Real-Time Telemetry

Continuous, high-precision logging of live Voltage (V) and Current (I) data.

2

Visual Status Mapping

Uses clear paths (Warning Path, System Processing, Operator Input) alongside dedicated LED indicators to give operators instantaneous system diagnostics.

3

Automated Load Shedding

The system can be programmed to automatically disconnect non-critical if total power consumption exceeds a safe threshold.

4

Robust Manual Override

Industrial-grade pushbuttons allow immediate manual control over any load, completely bypassing automated software routines if necessary.

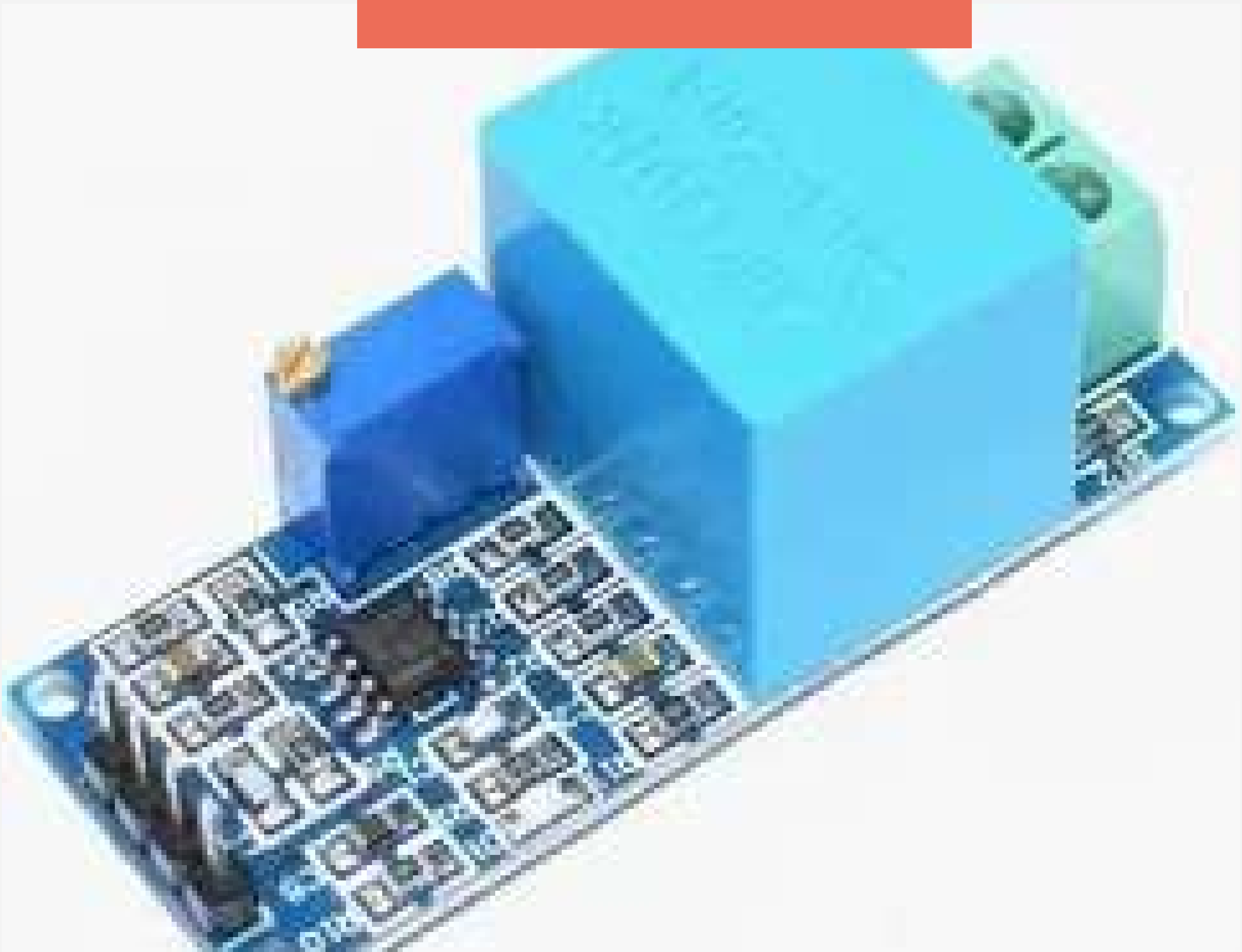
Processing Brain

Arduino Uno Controller:
Manages all system automation, sensor calculations, and output triggers.

Telemetry Array

Current & Voltage Sensors

Dedicated modules providing continuous analog feedback of electrical characteristics.



Power Switching

4-Channel Relay Array
Actuators that isolate the low-power control signals from the high-power industrial loads.

Hardware Protection

Circuit Breakers
Standard industrial-grade physical interruption devices for absolute overcurrent protection.

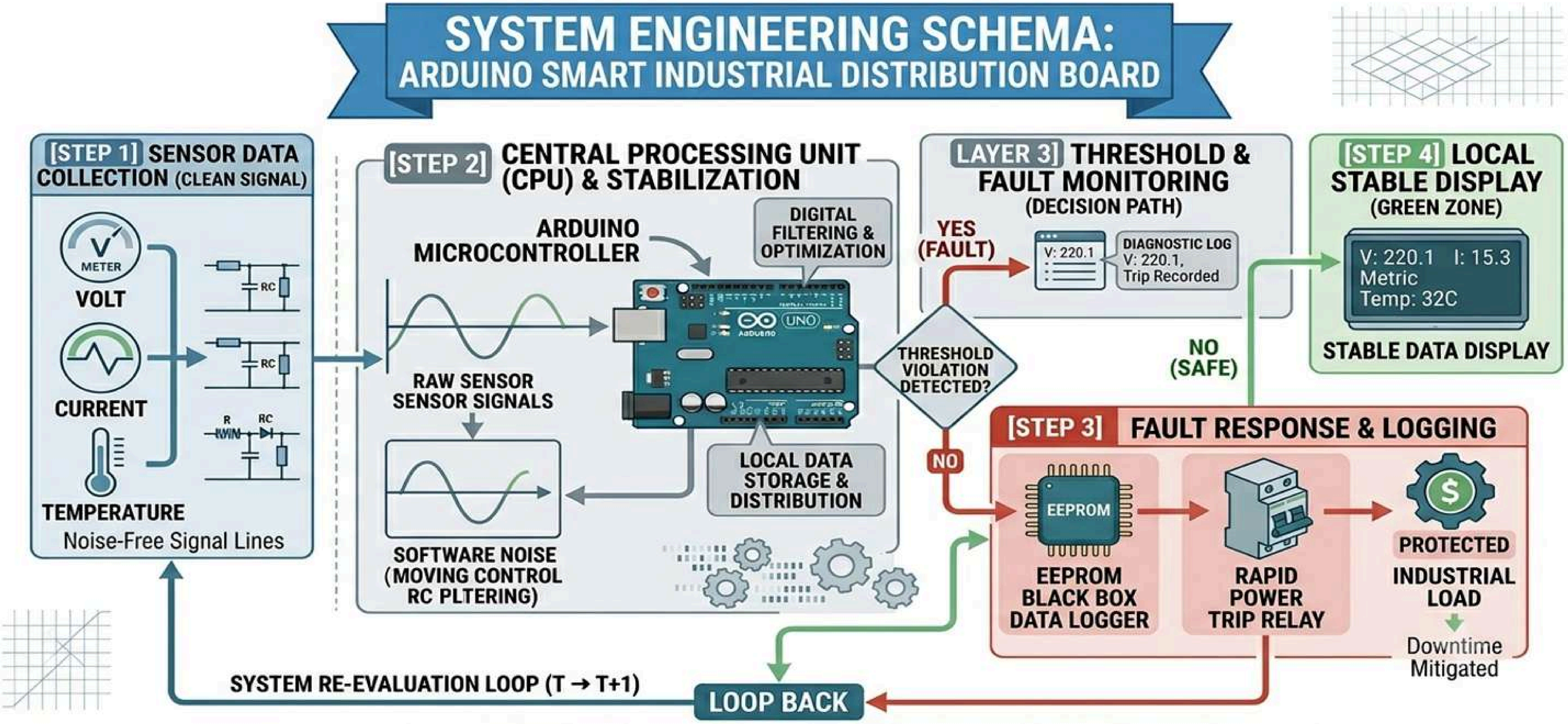
Power Conditioning

Voltage Regulator:
Steps down and stabilizes incoming power to safely feed the electronic control board.

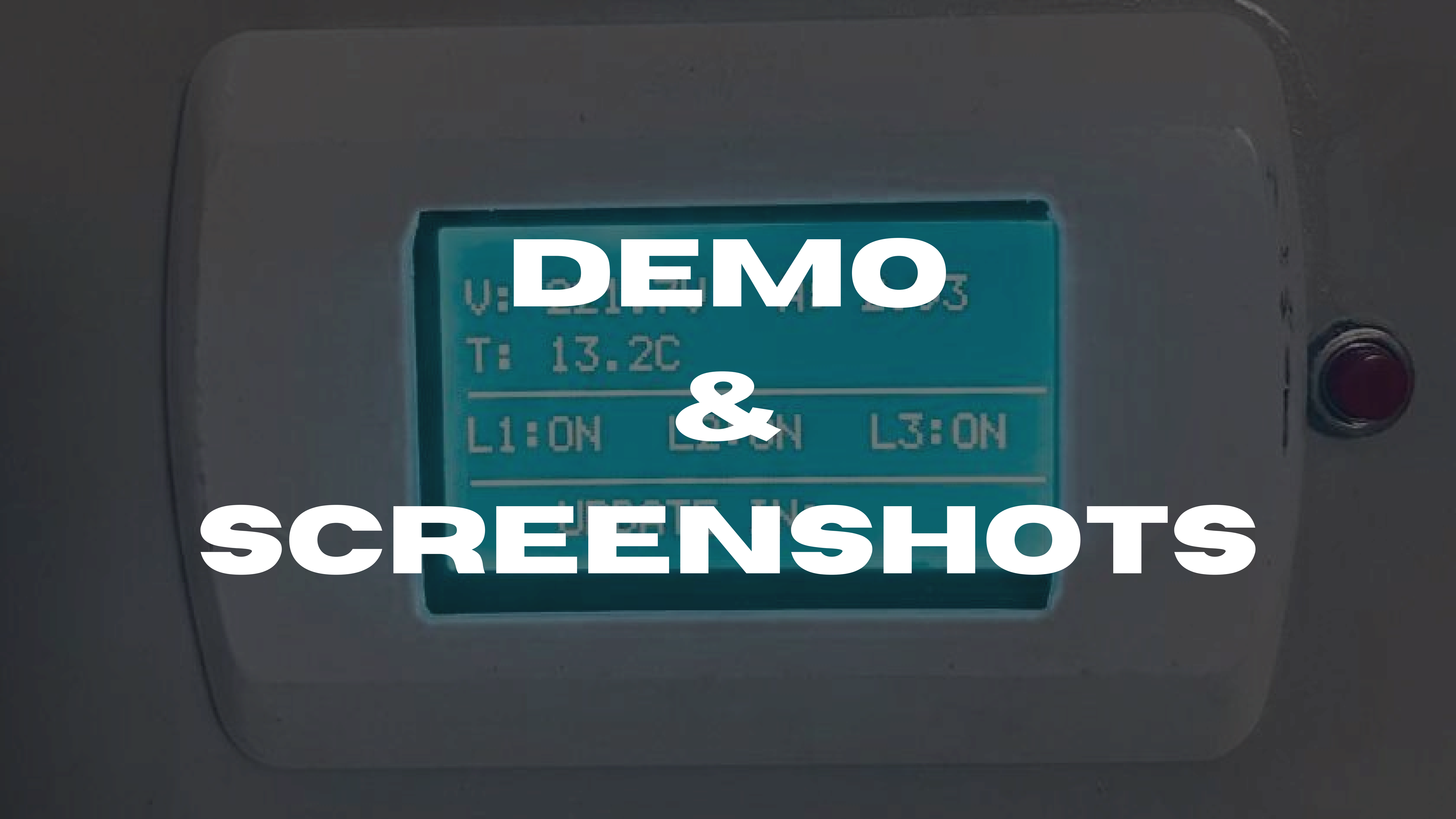
Cooling Fan
Drives active airflow across the internal components to maintain operational reliability.

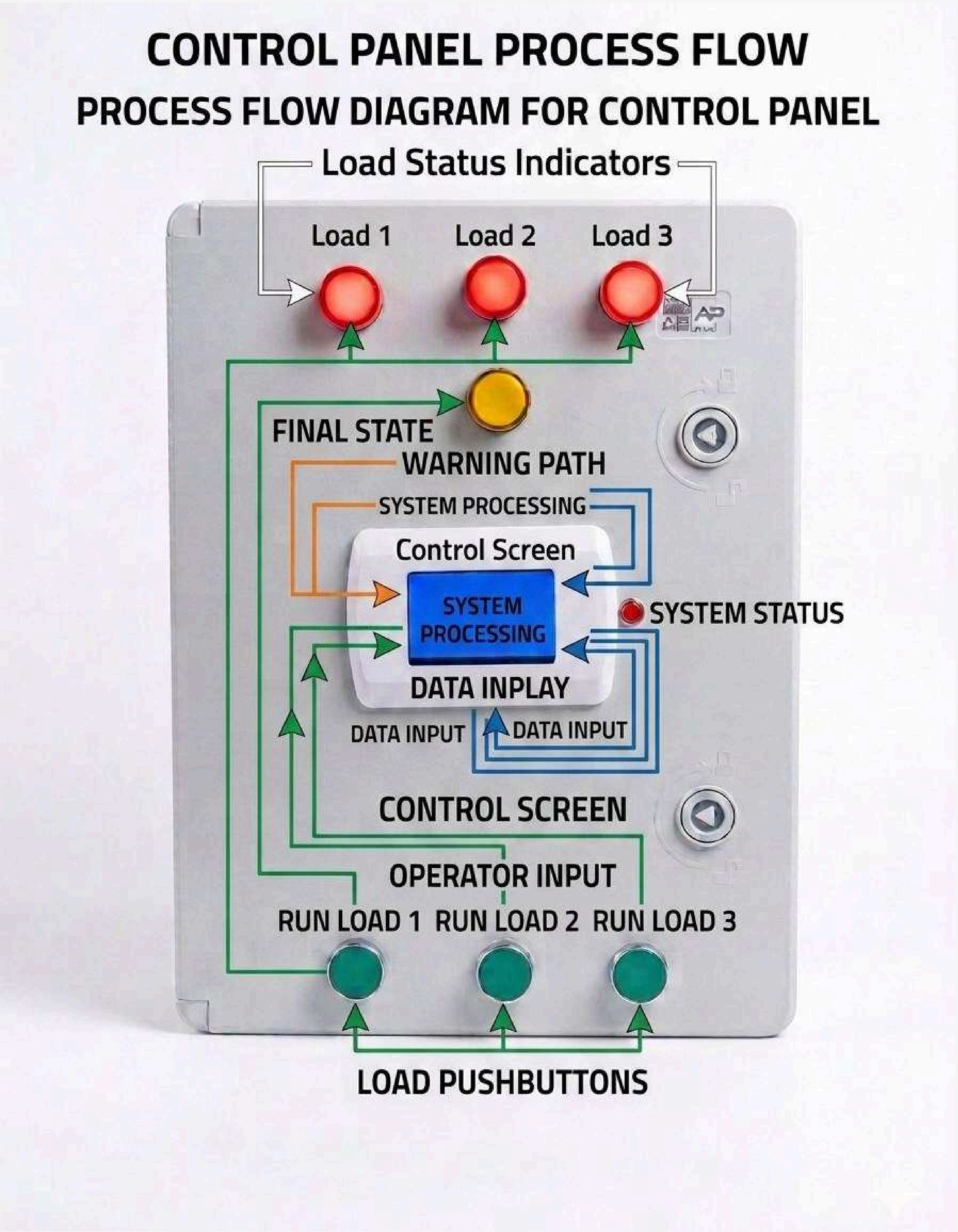
Thermal System

SYSTEM ARCHITECTURE



DEMO & SCREENSHOTS



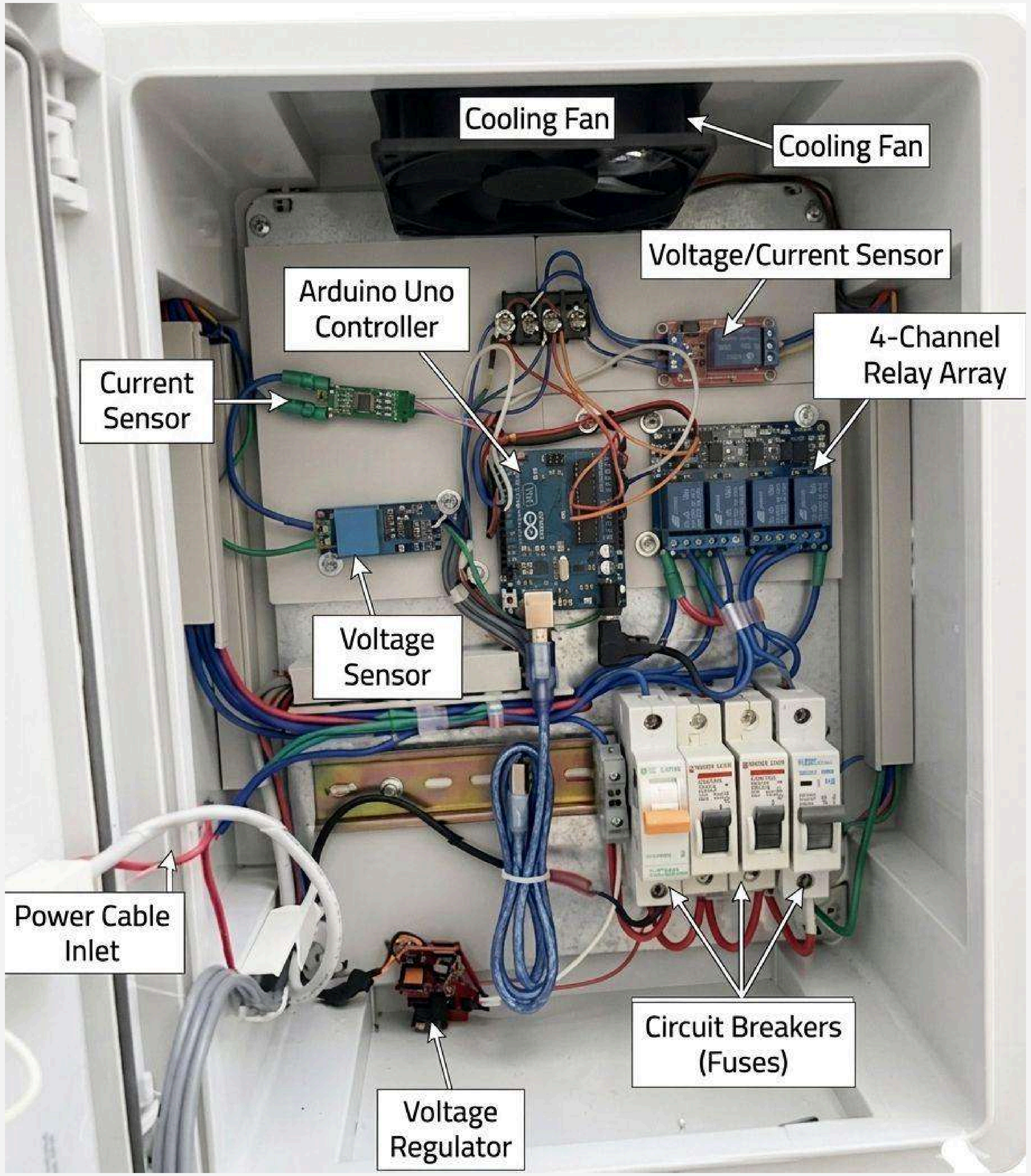


The project's exterior

This image describes the components on the project interface and explains their functions.

The project's interior

This image shows the basic components we used in the project.



Project display screen

The screen displays important and essential data such as current, voltage, panel temperature, and load status (whether on or off). If any system error occurs, the screen displays blackbox for the system

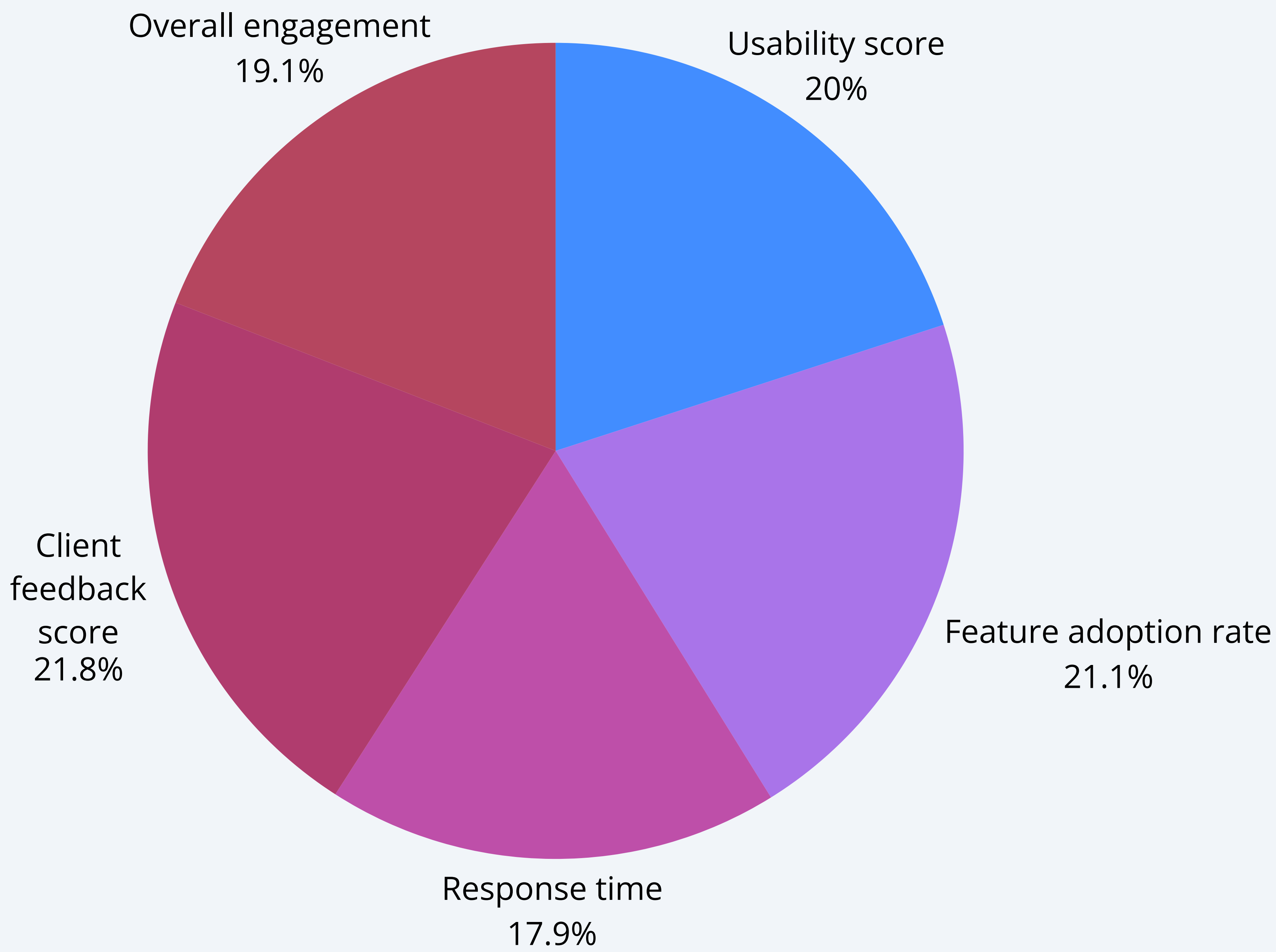
Results & Evaluation

1

91.3%
Accuracy / score

2

0.94
Performance metric



3

4.7/5
User Satisfaction

PROJECT

Timeline

WK 1-2	WK 3-4	WK 5-8	WK 9-10	WK 11-12
Research & Requirements	Design & Prototyping	Development Sprint	Testing & Evaluation	Final Report & Demo


THANK
YOU